1 Summary

Reaction/Process Name:			
Your name:		**Your demonstrator:**	
Date of experiment:		**Your laboratory:**	

Chemical inventory

Chemical	**Quantity**	**Hazardous?**									

2 Use

Describe the reaction or process, taking care to include any additional risks introduced (e.g. pressure, explosion hazard, gas evolution, exothermic reaction):

3 Accident or emergency

Turn off ☐ Power ☐ Water ☐ Vacuum ☐ Gas

In case of spillage or uncontrolled release:

In case of fire:

Treatment to be adopted if personnel are affected:

Student

Demonstrator

Robert Herman
Technical Operations Coordinator

4 Exposure routes

How can exposure occur?	Dermal (skin)	Eyes	Inhalation	Ingestion	Injection
Who is exposed?					

5 Potential health effects

Acute (immediate) effects:	
Chronic (delayed) effects:	

6 Required precautions and control measures

Ventilation controls:	
PPE controls:	
Other required controls:	
Required checks:	
Waste disposal:	

Hierarchy of controls:		1. Elimination
Tick only if control is		2. Substitution
implemented		3. Isolation
		4. Engineering
	✔	5. Administration*
**Mandatory control*	✔	6. PPE*

Health and safety risk matrix

Hover over each word in the matrix to the right to read about each descriptor.

- The risk rating is found at the intersection of each row (consequence) and column (likelihood).
- A control is any measure or action currently in existence that modifies or manages the risk, e.g. a policy, procedure, practice, process, technology, technique, method, or device.
- You will assess the risk rating by identifying the consequences and likelihood before and after controls are implemented — the risk rating should be the same or lower, or the control may not be effective.

		Likelihood level				
		Highly unlikely	**Unlikely**	**Possible**	**Likely**	**Highly likely**
Consequence level	**Catastrophic**	High	High	Extreme	Extreme	Extreme
	Major	Medium	High	High	Extreme	Extreme
	Moderate	Low	Medium	High	High	Extreme
	Minor	Low	Low	Medium	Medium	High
	Insignificant	Low	Low	Low	Medium	Medium

7 Risk ranking

Maximum foreseeable exposure
Use the health and safety risk matrix to determine the risk level **before** any controls in Section 6 are implemented:

Consequence level:	
Likelihood level:	
Risk level:	

Controls rating
Are you doing what is reasonable under the circumstances to prevent or minimise the impacts of the risk?

- ☐ **Excellent**
- ☐ **Adequate**
- ☐ **Inadequate**

Residual risk exposure
Use the health and safety risk matrix to determine the risk level **after** all controls in Section 6 are implemented:

Consequence level:	
Likelihood level:	
Risk level:	

Consider the following to develop an **overall risk ranking**:

- The route(s) of exposure, and who is exposed as detailed in Section 4;
- Any route(s) of exposure with the potential to cause harm as detailed in Section 5;
- Whether required control measures in Section 6 in place for all exposed individuals;
- Is the recommended PPE from the SDS being used?

Overall risk ranking:

Where an overall risk ranking of 3 or 4 is given, then additional controls will be necessary before work can commence.

By submitting this worksheet you agree to implement all required controls and to monitor their implementation to ensure you have a safe workplace while completing this experiment. You will only perform these experiments under the supervision of a Demonstrator. No unauthorised experiments will be undertaken.

Physical properties summary

Please provide a summary of the physical properties for each chemical in use during this experiment, as well as any specific disposal or clean up instructions. You will refer to this information during the laboratory.

Chemical	Quantity	Melting point	Boiling point	Flash point	Clean up

Student

Demonstrator

Robert Herman
Technical Operations Coordinator